PassFrame tuition

Angles & 2D Shapes

Year 4 Pt 2 — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: Three angles around a point are 100°, 130° and x°. What is x?

Method & answer: Angles around a point add to 360°. x = 360 − 100 − 130 = 130°.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What type of angle is 210°?
2. What type of angle is 82°?
3. Which of these is reflex? A) 45° B) 90° C) 135° D) 220°
4. What do angles on a straight line always add up to?
5. What do angles around a point always add up to?
6. How many sides does a heptagon have?
7. Which shape has 10 sides? A) octagon B) nonagon C) decagon D) heptagon
8. How many sides does a nonagon have?
9. Select ALL of these angles that are REFLEX (between 180° and 360°): A) 190° B) 120° C) 350° D) 60° E) 181°
10. The three angles inside any triangle always add up to how many degrees?
11. What type of triangle has exactly two sides equal?
12. Which shape has 5 lines of symmetry? A) square B) regular pentagon C) equilateral triangle D) rectangle
13. What type of triangle has all angles less than 90°?
14. What is the name for an angle less than 90°?
15. How many sides does a quadrilateral have?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Two angles on a straight line are 96° and x°. What is x?
2. Three angles around a point are 105°, 115° and x°. What is x?
3. A triangle has angles of 38° and 82°. What is the third angle? A) 50° B) 60° C) 70° D) 80°
6. In a triangle, two angles are equal at 62° each. What is the third angle?
7. Two straight lines cross, forming vertically opposite angles. If one angle is 51°, what is the angle vertically opposite it?
8. Select ALL of these that describe a rectangle: A) opposite sides are equal B) all 4 angles are 90° C) all 4 sides are equal D) opposite sides are parallel E) it has 4 lines of symmetry
11. A quadrilateral has angles of 100°, 70°, and two other equal angles. What is the size of each of the other two?
12. Which of these quadrilaterals always has 4 equal sides? A) rhombus B) rectangle C) trapezium D) parallelogram
13. The exterior angle of a triangle equals the sum of the two opposite interior angles. If those are 42° and 58°, what is the exterior angle?

4. A quadrilateral has angles of 80°, 95° and 105°. What is the fourth angle?

5. Two equal angles lie on a straight line. What is the size of each?

9. An isosceles triangle has a base angle of 62°. What are the other two angles?

10. A right angle is split into two angles, one of which is 33°. What is the other angle?

14. A triangle has one angle of 90° and another of 32°. What is the third angle?

15. The interior angles of a regular pentagon sum to 540°. What is each interior angle?

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Two angles on a straight line are (x+30)° and (2x)° . What is x, and what are the two angles?

2. Angles a and b lie on a straight line, and a = 3b. Find b. A) 30° B) 45° C) 60° D) 90°

3. A regular polygon has an exterior angle of 45°. How many sides does it have?

4. Select ALL of these that are TRUE about a trapezium: A) it has exactly one pair of parallel sides B) all 4 sides must be equal C) it is a quadrilateral D) it always has 2 right angles E) it has 4 sides

5. A ladder leans against a wall, making an angle of 62° with the ground. What angle does it make with the wall?

6. A pizza is cut into equal slices, each with a 36° angle at the centre. How many slices are there in total?

7. Two parallel lines are cut by a transversal, forming co-interior angles of 65° and x°. Find x.

8. A triangle has angles in the ratio 2:3:5. What is the size of each angle?

9. A quadrilateral's angles are all equal, and it also has 4 equal sides. What shape must it be?

10. A door is opened so it makes an angle of 48° with the wall. Through how many more degrees must it open to be at a right angle to the wall?

11. A regular polygon has an interior angle of 144°. How many sides does it have?

12. A triangular garden bed has one angle of 90° and another of 42°. What is the third angle, and what type of triangle is this?

13. Two straight roads cross at a junction, forming four angles. One angle measures 118°. What are the sizes of the other three angles?

14. A kite-shaped flag has two equal angles of 70° each and one other angle of 125°. What is the size of the fourth angle?

15. An isosceles triangle has an apex angle of 52°. What are the two base angles?